Nick Detection in Cricket via Pre- and Post-Impact Momentum Analysis

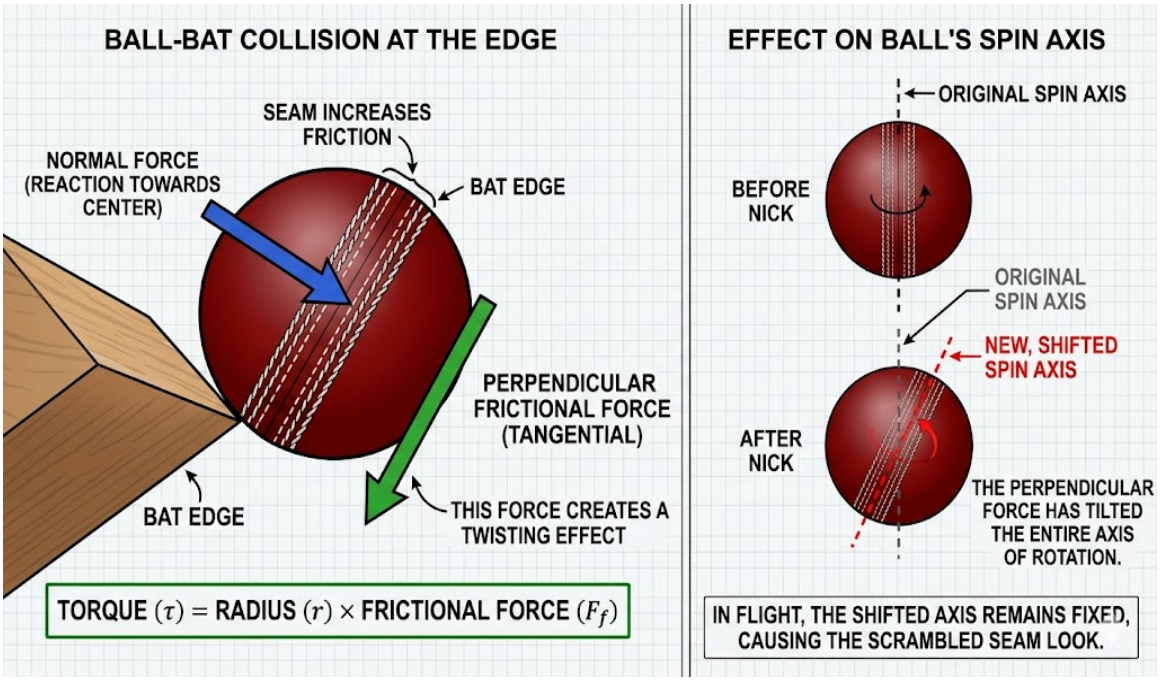


Figure: Spin axis change due to tangential force at the point of impact.

Introduction

In cricket, there are scenarios where the ball brushes the bat but the interaction is too subtle to be detected by existing methods such as audio-based systems or standard visual tracking.

To address this limitation, a physics-based approach can be used to determine whether a nick has occurred by analyzing changes in linear and angular momentum before and after the ball passes the bat.

Core Idea

A bat-ball interaction introduces a force impulse affecting linear momentum and a torque impulse affecting angular momentum.

Thus, even if traditional systems fail, momentum analysis can reveal hidden nicks.

1. Impact Mechanics of a Nick

Normal Force: Acts through the center and causes rebound.

Tangential Force: Acts along the surface and generates torque.

2. Effect on Ball Motion

Linear Motion: Change in speed and direction.

Rotational Motion: Change in spin rate, direction, and axis tilt.

3. Method 1: Linear Momentum Analysis

Compare motion before and after crossing the bat.

Detect changes in velocity magnitude and direction.

4. Method 2: Angular Momentum Analysis

A nick introduces torque causing change in angular momentum.

Axis passes through CM and is perpendicular to radius vector from point of impact.

5. Determining Spin

Equation: $v = \omega \times r$

v = velocity, r = position vector, ω = angular velocity

6. Practical Estimation Method

Capture high-speed video

Track seam points

Compute velocities

Solve for angular velocity ω

7. AI-Based Enhancement

Use computer vision and deep learning

Automate tracking and estimation

Works in real match conditions

8. Why This Method Works

Even without sound or visible deviation, momentum changes always occur.

These are fundamental signatures of a nick.

9. Conclusion

Momentum-based analysis is a robust method to detect subtle nicks.

It works where conventional systems fail.